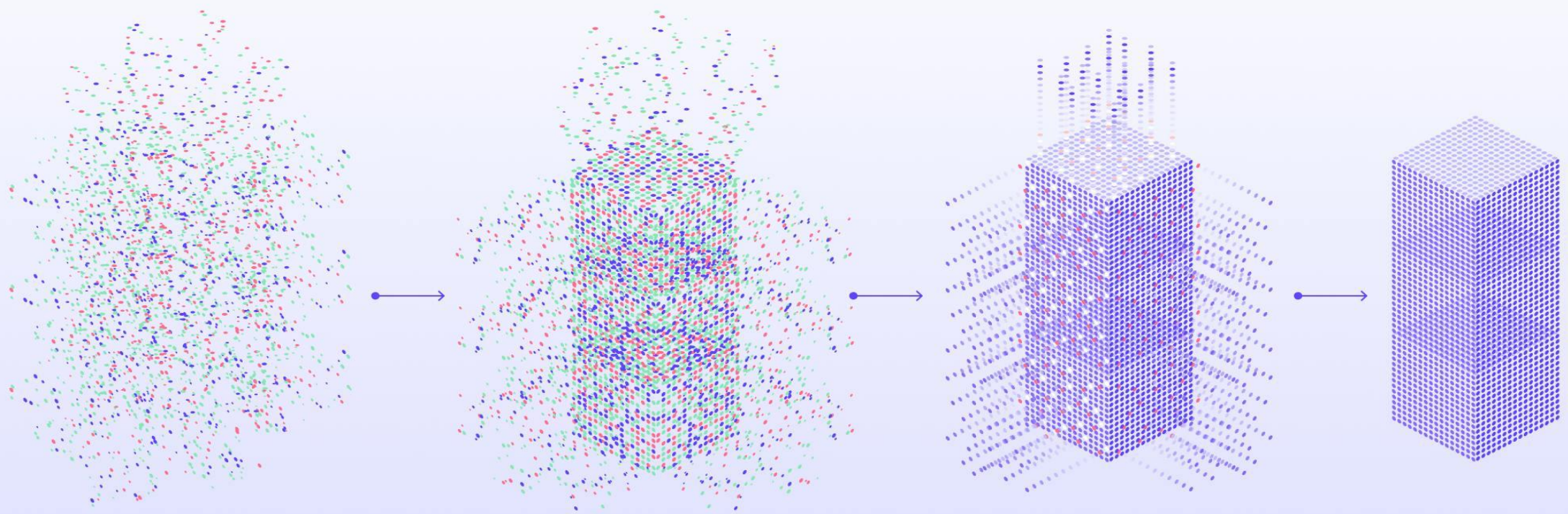


Definition of Data Visualization

- **Data Visualization** is the systematic process of representing data in graphical or visual formats—such as charts, graphs, plots, and dashboards—to enable efficient understanding, analysis, and communication of information.
- Data visualization serves as a bridge between **raw data and human cognition**, allowing patterns, trends, correlations, and anomalies to be perceived more quickly than through numerical tables alone.
- Visualization is a **downstream consumer of engineered data**, transforming cleaned, structured, and aggregated datasets into interpretable insights.
- The emphasize that visualization is not merely a presentation layer but a **critical feedback mechanism** in the data lifecycle, influencing business decisions and system improvements.







DIPHTHERIA.

SCROFULA.

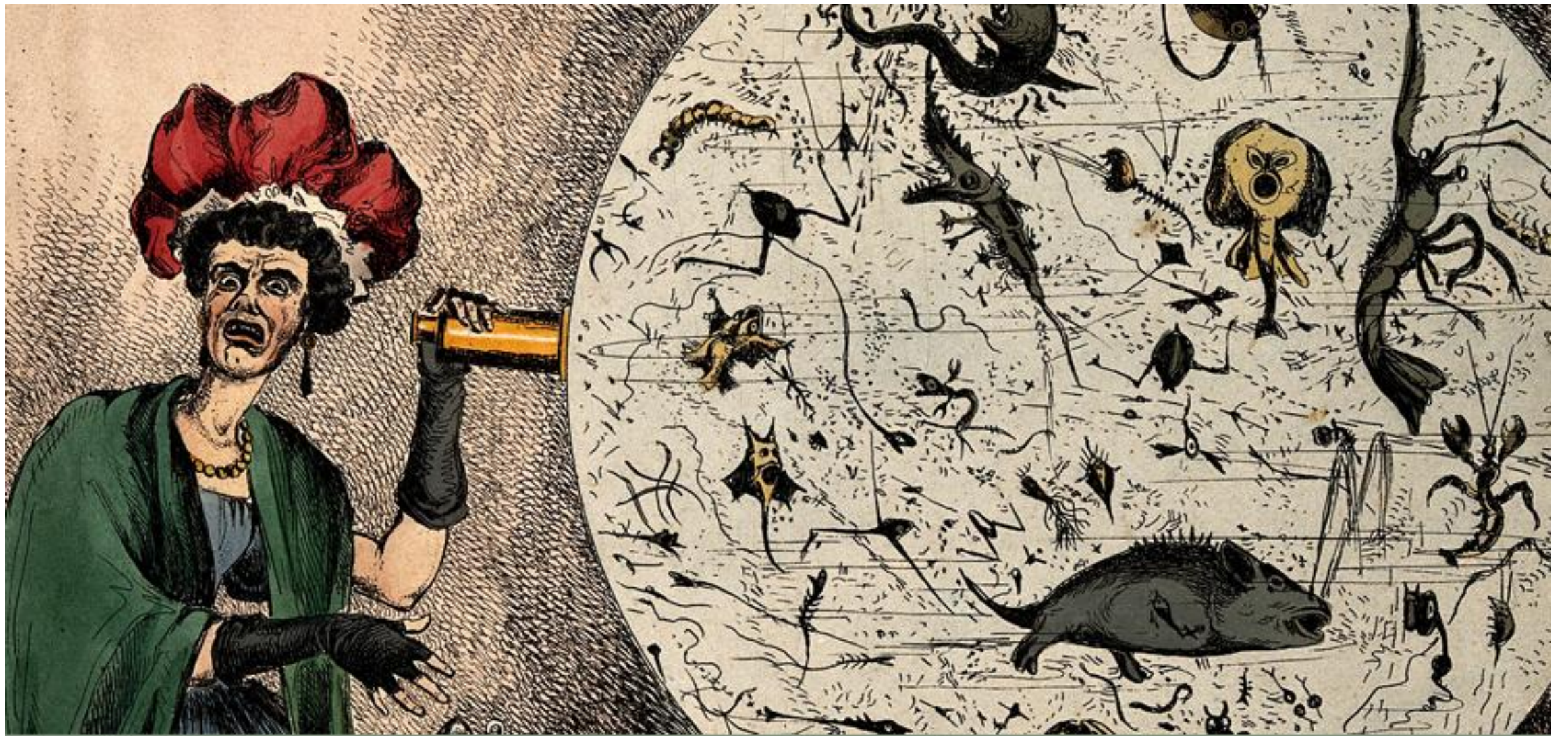
CHOLERA.

FATHER THAMES INTRODUCING HIS OFFSPRING TO THE FAIR CITY OF LONDON

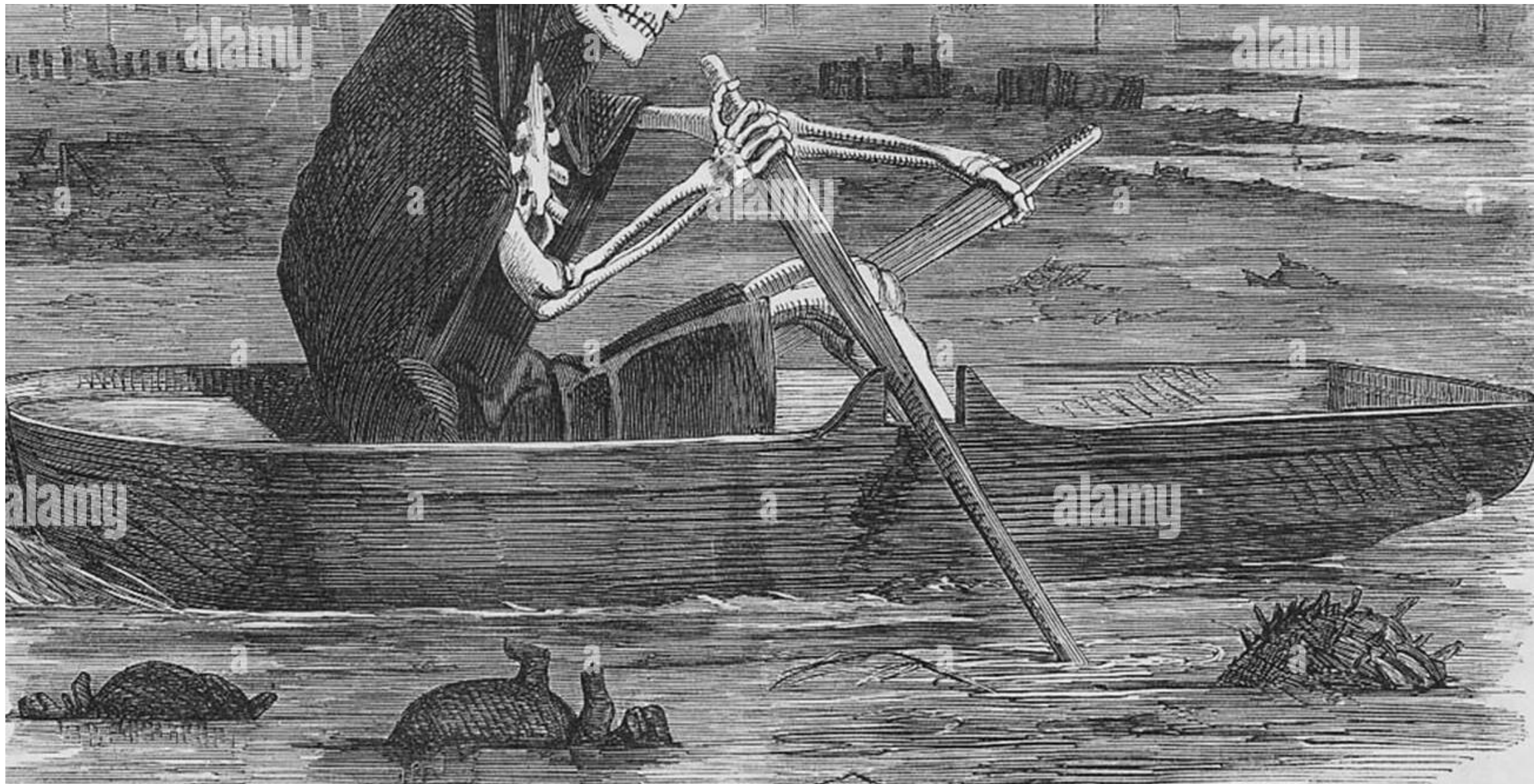
(A Design for a Fresco in the New Houses of Parliament.)



CHOLERA "TRAMPLES THE VICTOR & THE VANQUISHED BOTH."



Credit: Wellcome Library, London. Wellcome Images - images@wellcome.ac.uk

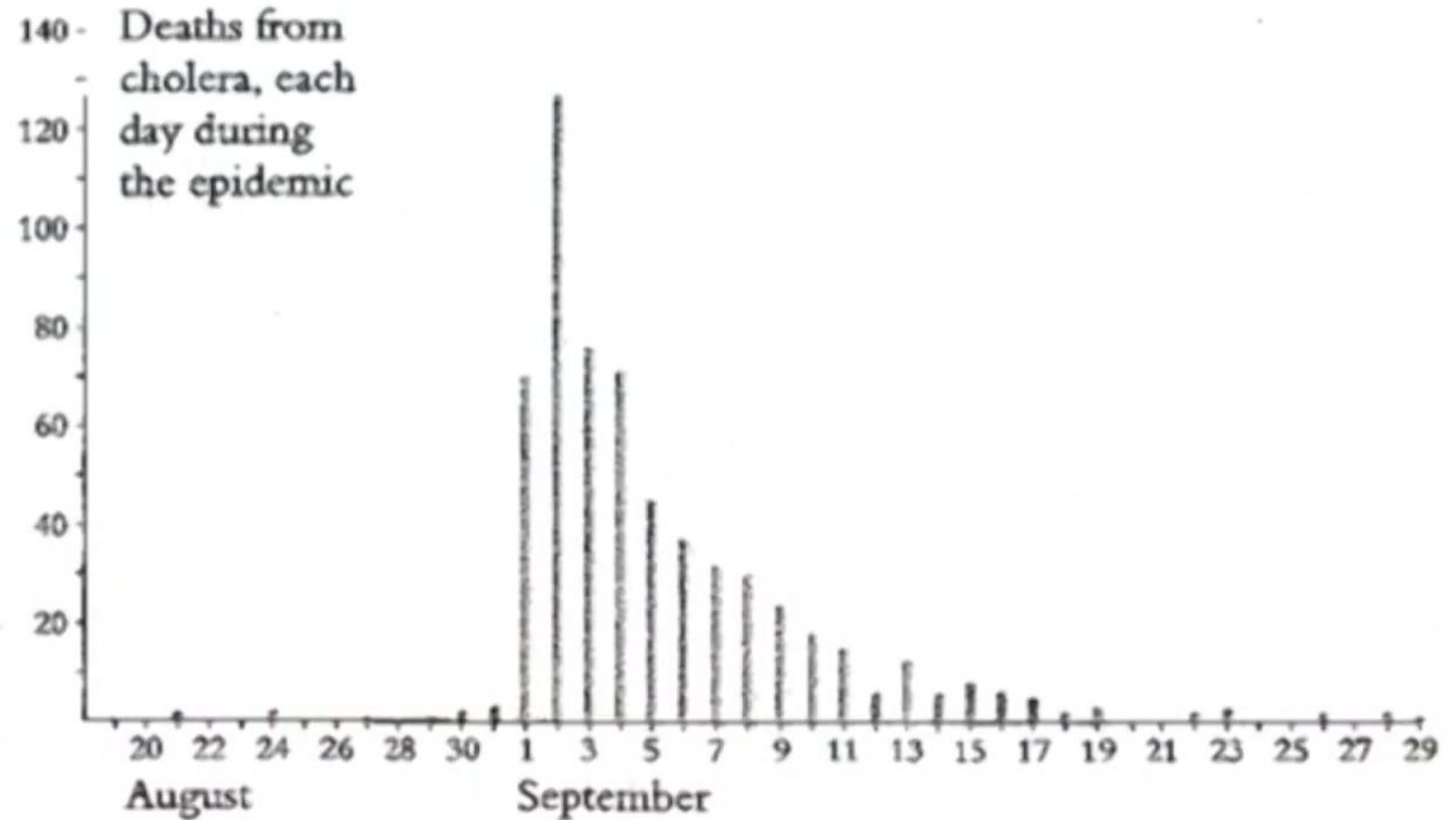


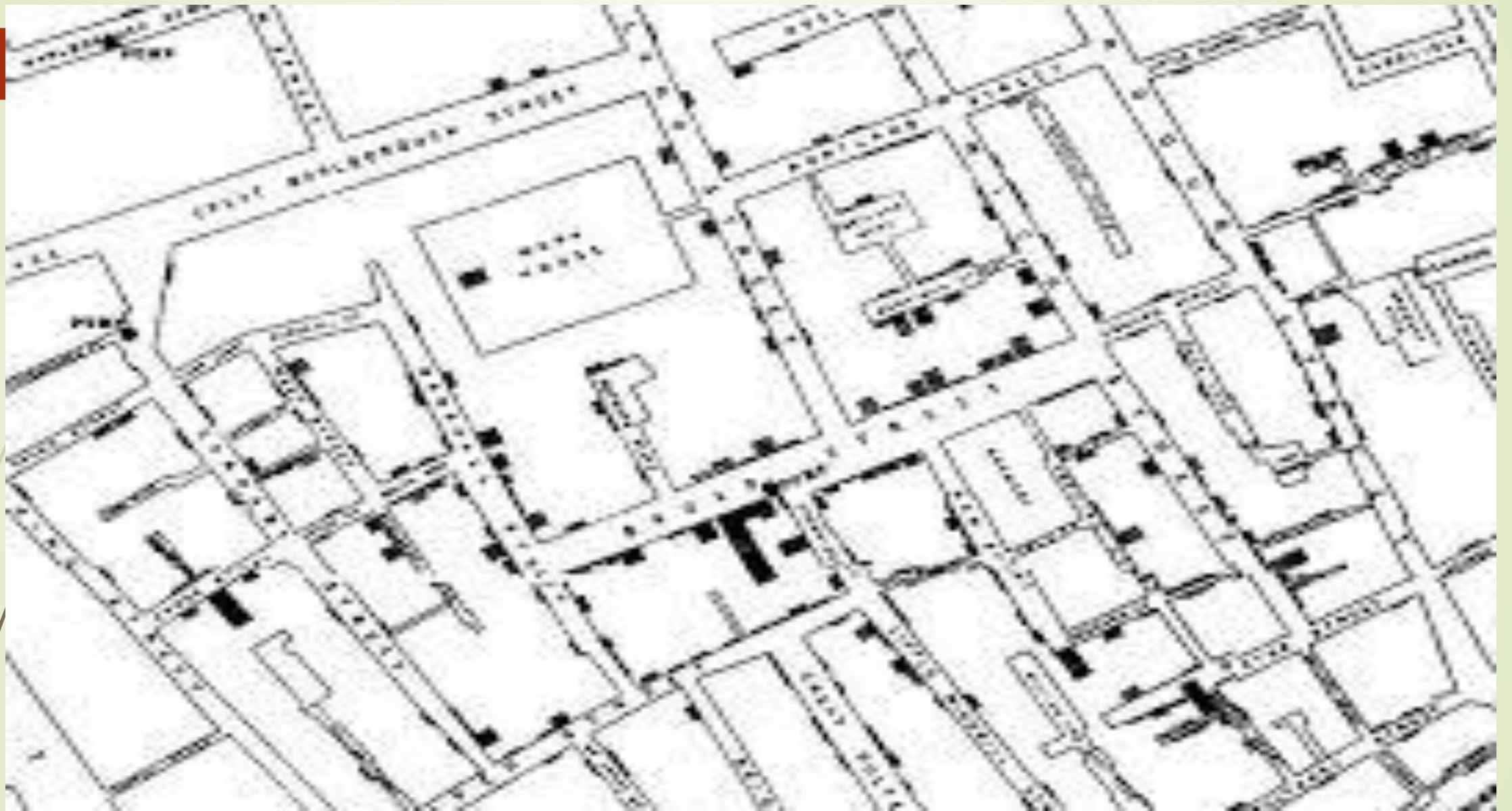


DEATH'S DISPENSARY.

OPEN TO THE POOR, GRATIS, BY PERMISSION OF THE PARISH.

Cholera epidemic 1854 London – Dr Snow





Visualization typology:-

1. **Geospatial** relate to real life physical locations, overlaying familiar maps with different data points.

➤ Flow map (e.g :- [Minard's Napoleon's March](#))

➤ [Density map](#)

➤ Cartogram(https://en.wikipedia.org/wiki/File:Global_population_cartogram.png)

➤ Heat map

(https://upload.wikimedia.org/wikipedia/commons/thumb/e/e7/World_heat_map.png/2560px-World_heat_map.png)

Classification based on the form

2. Network Datasets connect deeply with other datasets. Network data visualizations show how they relate to one another within a network. In other words, demonstrating relationships between datasets without wordy explanations.

- Matrix charts
- Node-link diagrams
- Word clouds



Classification based on the form

3. Temporal data visualizations track time series data - the performance of an indicator over a period of time. They normally feature lines that either stand alone or overlap with each other, with a start and finish time.

- Line graph, bar chart, gantt chart, area chart, scatter plots
- Polar area diagrams (Coxcomb chart, Florence Nightingale)
- Time series sequences
- Timelines
- Line graphs



Classification based on the form

Hierarchical data visualizations order groups within larger groups. They are best suited if you're looking to display clusters of information, especially if they flow from a single origin point.

- Tree diagrams
 - Ring charts
 - Sunburst diagrams
- 

Importance of Data Visualization

1. Enhancing Human Understanding of Data

- Humans process visual information faster than textual or numerical data. Visualization exploits **pre-attentive attributes** (color, size, position) to allow immediate recognition of trends and outliers.
- Explains that effective visuals reduce cognitive load by guiding the viewer's attention to what matters most.
- **They** demonstrates how visual exploration is often the first step in understanding a dataset before applying statistical models.

Real-life

example:

In an e-commerce platform, daily sales data across thousands of products is difficult to interpret in tabular form. A **line chart showing revenue trends** instantly reveals seasonal peaks or sudden drops caused by system outages or promotional campaigns.



2. Supporting Decision-Making in Data-Driven Systems

In *Fundamentals of Data Engineering*, visualization is described as a **decision interface**—the final stage where data informs action.

Visualization enables:

- Faster business decisions
- Evidence-based policy formulation
- Monitoring of KPIs (Key Performance Indicators)

Real-life example:

A logistics company uses a **geographical heat map** to visualize delivery delays across regions. Decision-makers can immediately identify bottleneck areas and allocate resources accordingly.



3. Detecting Patterns, Trends, and Anomalies

Visualization allows analysts to detect:

- Long-term trends
- Short-term fluctuations
- Unexpected anomalies

Highlights visualization as an essential tool for **observability in large-scale systems**, helping engineers detect failures or performance degradation.

Real-life

example:

A system reliability team visualizes server response times. A **sudden spike in latency** on a time-series chart reveals a database bottleneck before it causes a system crash.



Communicating Insights to Non-Technical Stakeholders

Raw data or code-based analysis is often inaccessible to managers or policy-makers.
Visualization acts as a **common language**.

The stresses that dashboards built on dimensional models enable executives to interact with data intuitively without understanding database schemas.

Real-life example:

A university administration dashboard displays **student enrollment trends** using bar charts and pie charts, allowing non-technical administrators to plan faculty hiring and infrastructure expansion.



Validation of Data Quality and Data Pipelines

As per **Paul Crickard**, visualization is frequently used by data engineers to:

- Validate ETL outputs
- Detect missing values
- Identify skewed distributions

Real-life example:

After ingesting sensor data from IoT devices, engineers plot a **histogram of readings**.
An unexpected flat distribution indicates faulty sensors or ingestion errors.

Principles of Effective Data Visualization

1. Clarity and Simplicity

Effective visualization should prioritize **clarity over decoration**.

- Avoid unnecessary colors, gridlines, and 3D effects.
- Every visual element should serve a purpose.

To emphasizes “removing clutter” to focus attention on the message.

Real-life example:

Instead of a complex 3D pie chart showing department-wise expenses, a **simple horizontal bar chart** clearly compares spending across departments .

2. Accuracy and Integrity

Visualization must represent data truthfully without distortion.

- Scales should not be misleading.
- Proportions must reflect actual values.

VanderPlas warns against truncated axes that exaggerate differences.

Real-life example:

In election data visualization, using a **zero-based y-axis** ensures that vote share differences are not visually exaggerated.

3. Appropriate Choice of Visual Encoding

Different data types require different visual forms.

- Categorical data → Bar charts
- Time-series data → Line charts
- Distribution → Histograms, box plots

Real-life example:

Using a **line chart** to show monthly electricity consumption is more effective than a pie chart, as it highlights temporal trends.

4. Context and Meaning

Data without context can be misleading.

- Titles, labels, legends, and annotations are essential.
- Comparisons should be meaningful.

Supports visualization by providing consistent business context through facts and dimensions.

Real-life example:

A chart showing “Revenue Growth” becomes meaningful only when annotated with **market expansion or pricing changes**.



5. Audience-Centric Design

Visualization should be designed according to the audience's expertise and needs.

- Executives → High-level dashboards
- Analysts → Detailed exploratory visuals
- Engineers → Diagnostic plots

Stress aligning visualization with **business use cases**.

Real-life example:

A CEO dashboard focuses on **KPIs and trends**, while a data scientist dashboard includes **scatter plots and residual plots** for deeper analysis.



6. Storytelling and Narrative Flow

Visualization should guide the viewer through a logical narrative.

To introduces the concept of **storytelling with data**, where visuals are arranged to support a clear message.

Real-life example:

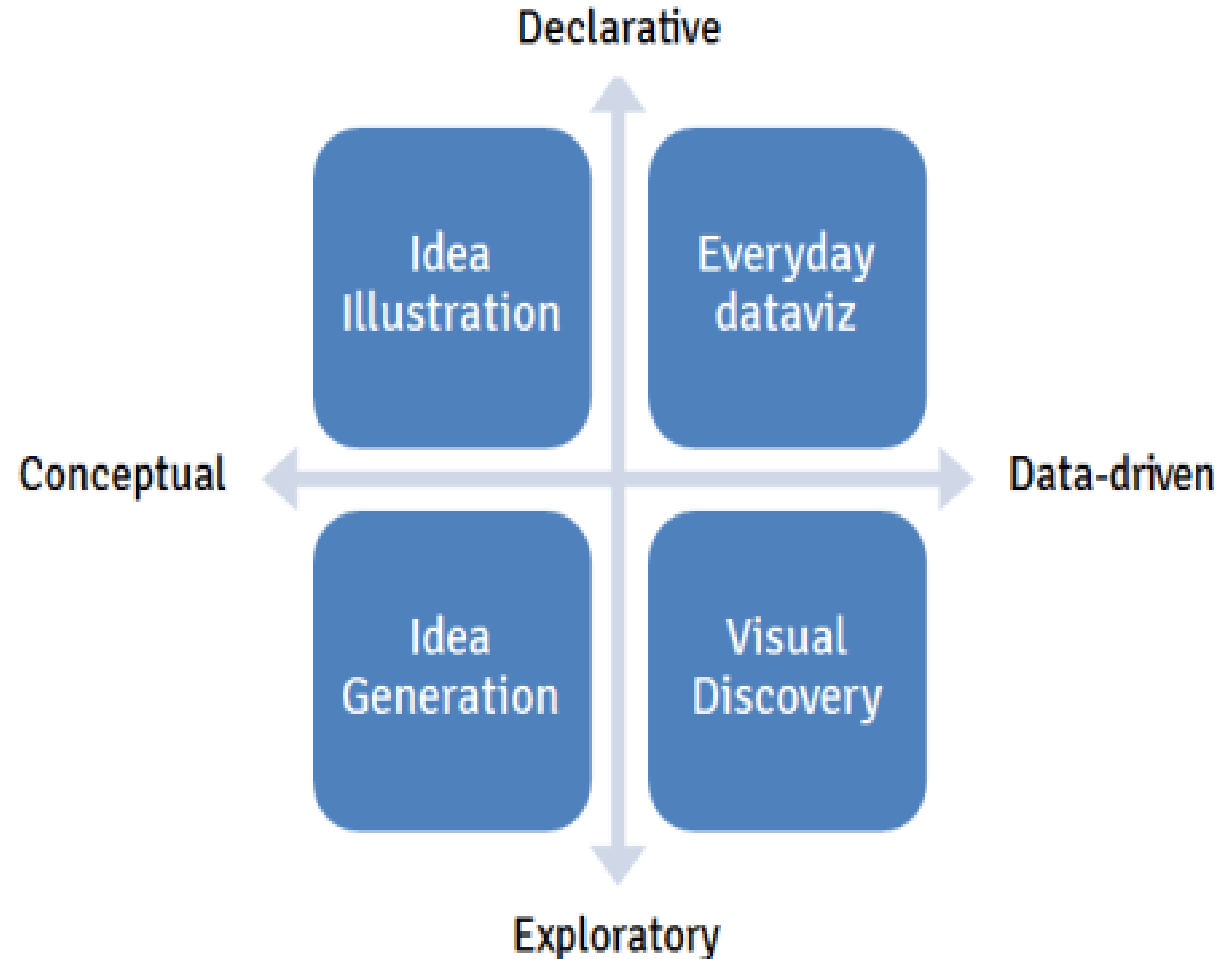
A public health report visualizes:

1. Rise in infection rates
2. Regional spread
3. Impact of interventions

This sequential narrative helps policymakers understand cause and effect.

The 4 types of Data Visualization:

Scott Berinato, HBR, June 2016



1. Idea Illustration

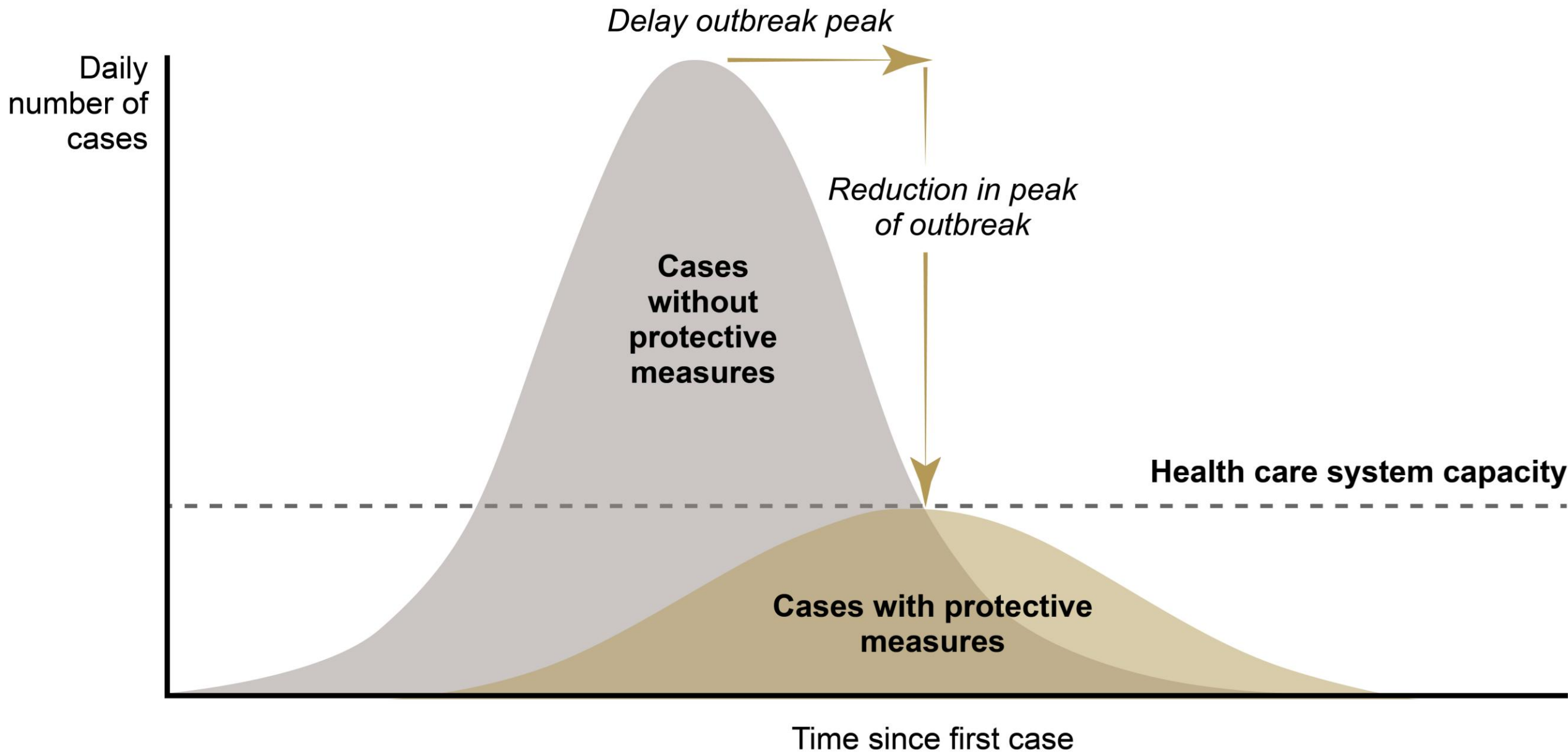
Idea illustrations clarify complex ideas by drawing on our ability to understand directions (up, down, forward, backward), metaphors (trees, bridges) and simple design conventions (circles, hierarchies). Org charts and decision trees are classic examples of idea illustration.

INFO TYPE: Process, framework

TYPICAL SETTING : Presentations, teaching

PRIMARY SKILLS : Design, editing

GOALS: Learning, simplifying, explaining



2. Idea Generation

Visualization as a tool to support idea generation. Like idea illustration, idea generation relies on conceptual metaphors, but it takes place in more-informal settings, such as off-sites, strategy sessions, and early-phase innovation projects.

INFO TYPE: Complex, undefined

TYPICAL SETTING: Working session, brainstorming

PRIMARY SKILLS: Team-building, facilitation

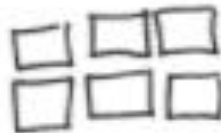
GOALS: Problem solving, discovery, innovation

①

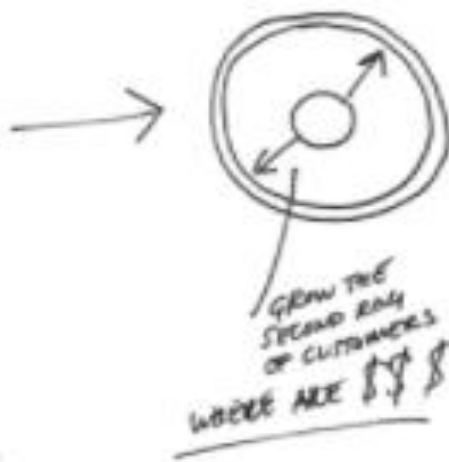
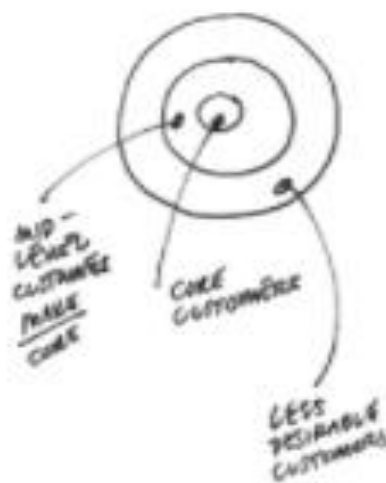
LOTS OF CUSTOMERS
SPENDING A LITTLE → FEWER CUSTOMERS
SPENDING MORE!



20 @ \$20 PER CUSTOMER
= \$2,400



6 @ \$500
= \$3,000



②

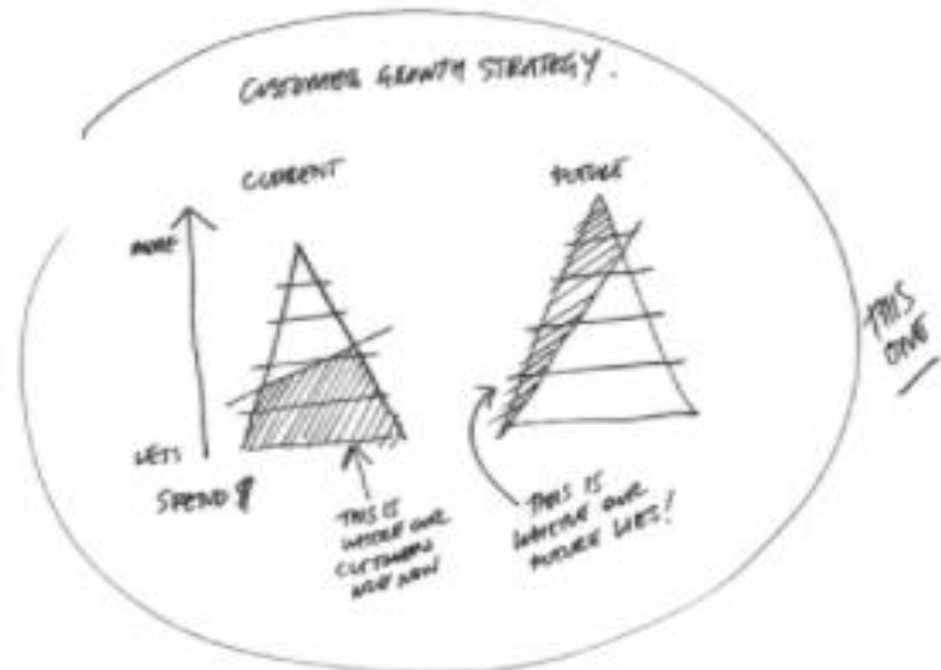
US NOW



FUTURE?



BIGGER
PIE.
FEWER
SLICES.?



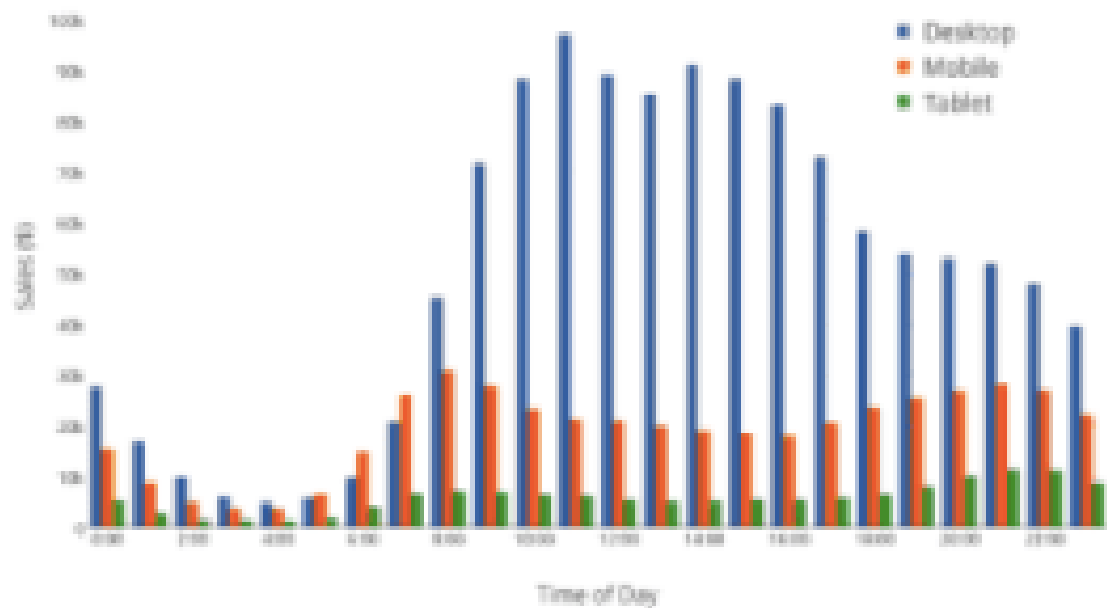
Everyday DataViz

Data-Driven Exploratory Visualizations displays complicated intricate data, with very involved visuals. Additionally, this visualization type is usually performed in a formal business setting with an end goal of spotting trends among data.

Scatter plots and line graphs are commonly used with this method as depicted in the image below.

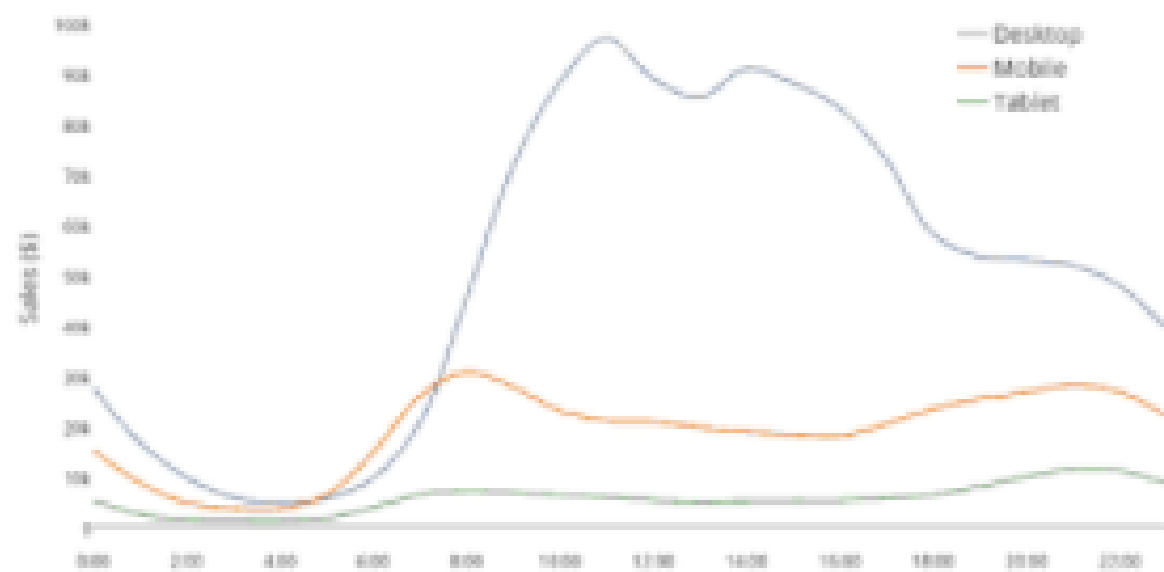
1

Sales by Time of Day



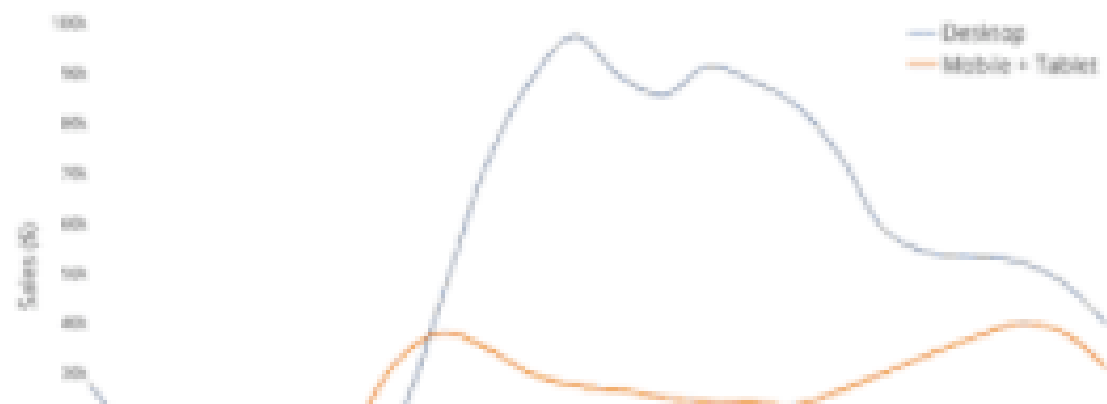
2

Sales by Time of Day



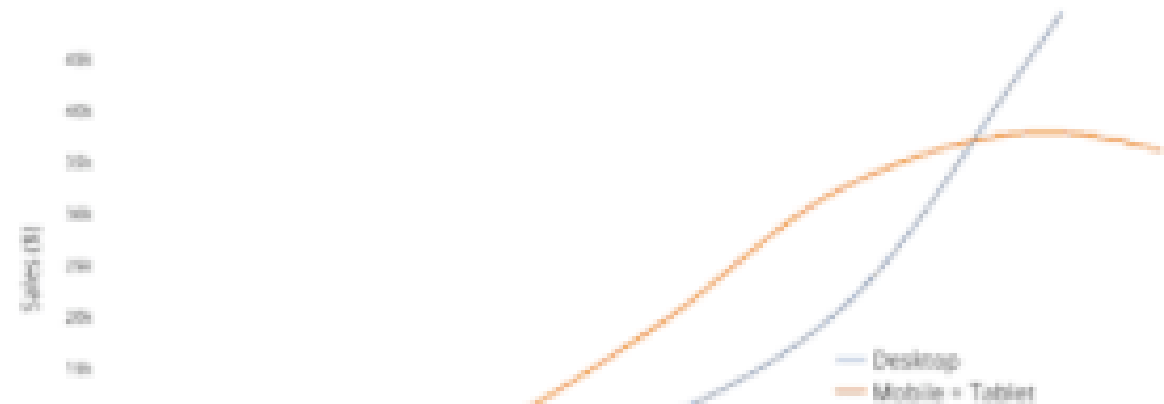
3

Sales by Time of Day



4

Mobile's Time of Day





4. Visual Exploration

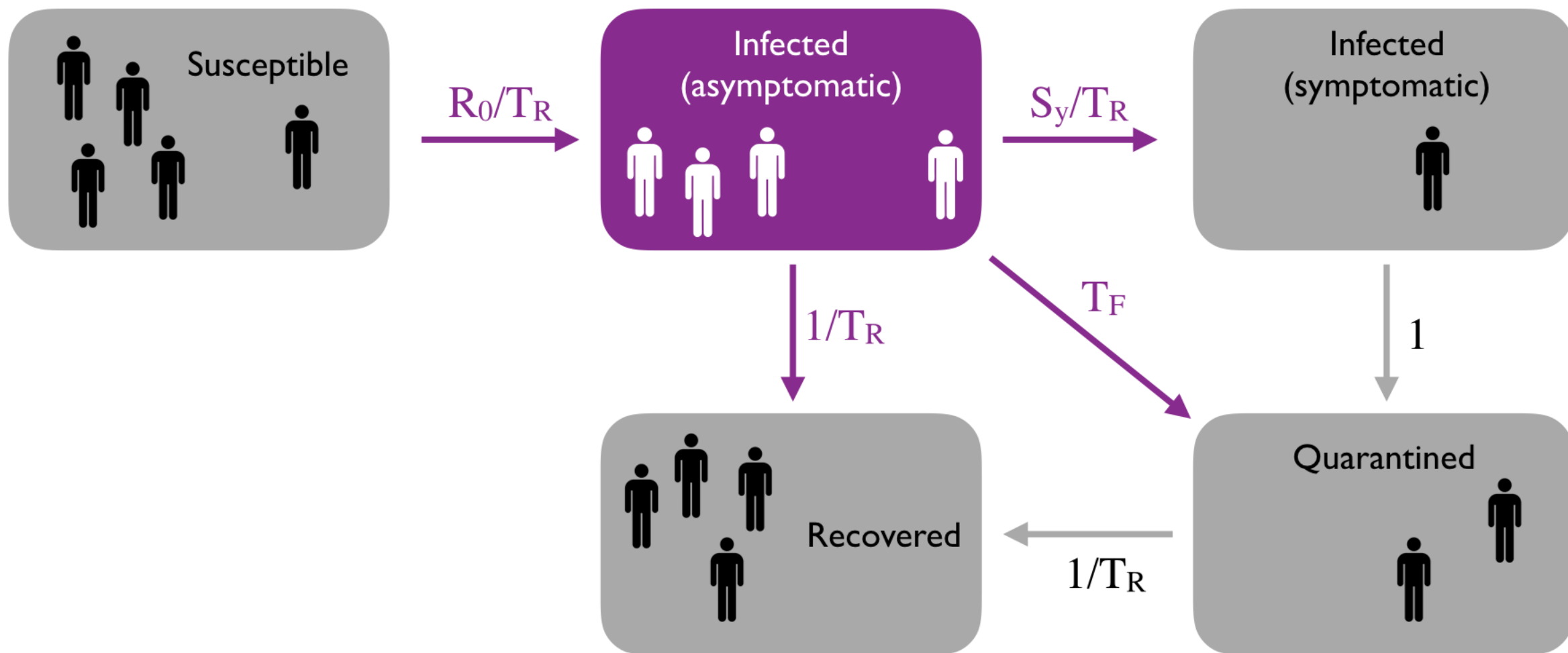
Open-ended data-driven visualizations It is intended to produces insights that can't be gleaned any other way.

INFO TYPE: Big data, complex, dynamic

TYPICAL SETTING: Working sessions, testing, analysis

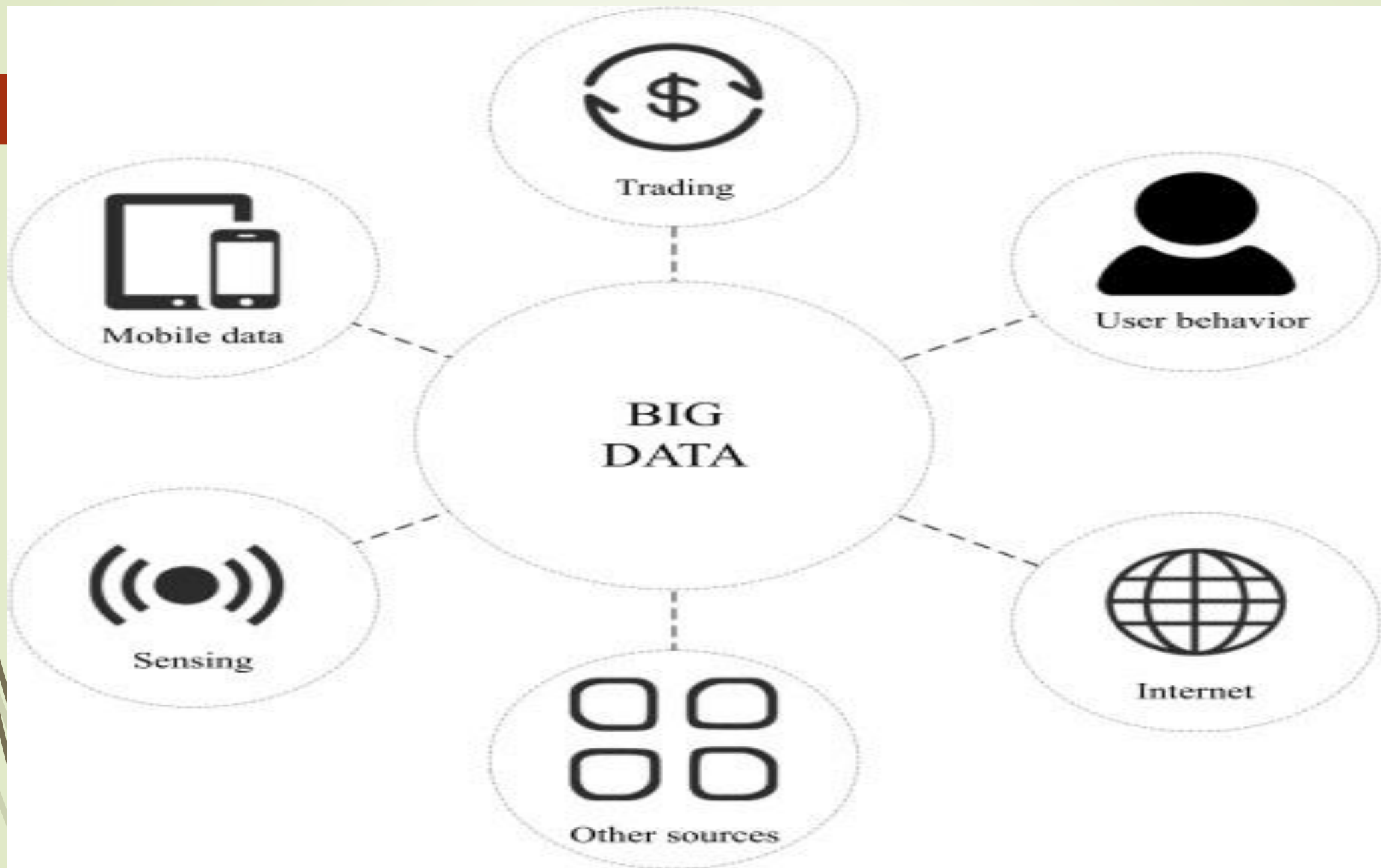
PRIMARY SKILLS: Business intelligence, programming, paired analysis

GOALS: Trend spotting, sense making, deep analysis



Definition of Data Generation

- **Data Generation** refers to the **process of creating data using computer software and algorithms**, typically to produce datasets with specific characteristics or within defined parameter ranges for analysis or experimentation.
- It includes the development of **large volumes of data points** — real or synthetic — that simulate real-world conditions, enabling testing, modeling, and evaluation in the absence of readily accessible actual data.
- In broader computing contexts (e.g., big data systems), data generation also refers to **how data originates from various sources** such as user activity, sensors, and logs.



Purpose and Context:-

- **Foundation of Data Workflows:** Data generation is often the **first stage in the data lifecycle**, supplying datasets that will be collected, stored, processed, and analyzed in subsequent steps.
- **Overcoming Data Scarcity:** It addresses the lack of sufficient or accessible real data — particularly important for machine learning and testing algorithms.
- **Controlled Experiments:** Generated data helps researchers and engineers **evaluate algorithms and systems under controlled conditions**.
- **Benchmarking and Validation:** In software testing and big data benchmarking, generated data serves as a standardized input to **assess system performance**.

Types of Data Generation

Data generation can be broadly classified according to the **method or purpose** used in practice.

A. Real Data Generation

- Occurs when systems or users in real environments produce data naturally, for example:
 - Website logs
 - IoT sensor outputs
 - Financial transactions
- This type forms the **basis for big data systems**, where massive streams of genuine activity become the raw material for analytics.

B. Synthetic Data Generation

- **Synthetic data** is artificially created and designed to resemble characteristics of real data without containing actual sensitive information.
- Synthetic data is generated via:
 - **Statistical models**
 - **Simulations**
 - **Generative learning models (e.g., GANs)**
- Useful in scenarios where data is:
 - Sensitive (privacy concerns)
 - Scarce
 - Expensive or impractical to collect
- Synthetic data includes **fully artificial datasets** as well as **augmented datasets** produced by transformation or expansion of existing data.

Common Methods for Data Generation:-

➤ **A. Rule-Based or Knowledge-Driven**

- Data is generated using **prescribed rules or domain expertise**.
- Useful when underlying processes are well-understood and can be formalized.

➤ **B. Data-Driven Methods**

- Use existing real data to learn patterns and distributions.
- Generative models (e.g., GANs, Bayesian nets) sample new data based on learned characteristics.

➤ **C. Hybrid Methods**

- Combine domain rules with real data characteristics to improve realism and relevance.
- Provide a balance between control and fidelity.

➤ **D. Simulation-Based Approaches**

- Simulate entire systems (e.g., discrete event simulation, agent-based models).
- Produce time series or process data resembling real system behaviors.

➤ Key Concepts and Characteristics

A. Parameterization

- Data generation typically defines **input variables and ranges**, ensuring that output covers **expected operational conditions**.

➤ B. Randomness and Diversity

- Many methods incorporate randomness to model **variation as seen in real environments**, influencing timing, sequence, and values across data streams.

➤ C. Quality Dimensions

- Generated data quality depends on:

- **Volume**

- **Variety**

- **Veracity (truthfulness)**

- **Velocity (rate of generation)**

- These factors influence the **utility and reliability** of generated datasets for downstream tasks.

Data Collection:-

- **Data collection** is the process of **gathering raw information from various sources** for use in analytics, reporting, and decision support.
 - It involves capturing data from **real-world events, user interactions, sensors, machines, logs, and external sources.**
 - **Objectives of Data Collection :-**
 - To acquire **accurate and relevant data** for analysis.
 - To support **data-driven decision making.**
 - To enable **learning and modeling** in AI/ML systems.
 - To gather data for **monitoring and control** of systems.
 - To feed **data pipelines and storage systems.**
- 

Data Collection Sources:-

A. Primary Data Sources

- Data collected directly from origin
- Examples:
 - Sensor readings (IoT devices)
 - User input in applications
 - Logs from servers and systems
- **B. Secondary Data Sources**
- Data collected from existing repositories
- Examples:
 - Data from public datasets
 - Data from third-party vendors
 - Archived logs

Methods of Data Collection:-

A. Manual Collection

- Human-driven processes
- Forms, surveys, observations
- High accuracy for specific contexts

B. Automated Collection

- Collected via software, sensors, and scripts
- Examples:
 - Application logs
 - API streams
 - IoT sensor data

Characteristics of Good Data Collection :-

➤ Accuracy

Data must reflect real conditions

➤ Completeness

No missing critical values

➤ Timeliness

Data must be collected when relevant

➤ Consistency

Same measurement standards over time

➤ Validity

Conforms to defined formats and rules

Challenges in Data Collection:-

➤ **Volume and Velocity**

Data generated rapidly from devices and users

➤ **Noise and Errors**

Unclean or incorrect data due to hardware/software issues

➤ **Heterogeneity**

Different formats, structures, and sources

➤ **Privacy and Security**

Sensitive data requires protection

➤ **Data Silos**

Fragmented data in multiple systems



Data Storage refers to the process of **systematically storing digital data** in physical or logical storage media so that it can be **retrieved, processed, and analyzed later**.

- The data storage is viewed as a **core component of the data lifecycle**, following data generation and preceding data processing and analysis.
- It involves:
 - Storage devices
 - Storage architectures
 - Data organization methods
 - Access and retrieval mechanisms



Objectives of Data Storage

- To **preserve data persistently**
 - To ensure **efficient retrieval**
 - To maintain **data integrity and consistency**
 - To support **scalability and availability**
 - To enable **secure access** to data
- 

Types of Data Storage:-

4.1 Primary Storage

- Also called **main memory**
- Includes RAM and cache
- Characteristics:
 - Fast access
 - Volatile (data lost when power is off)
 - Limited capacity

Secondary Storage

Used for **permanent data storage**

- Includes:
 - Hard disk drives (HDD)
 - Solid state drives (SSD)
- Characteristics:
 - Non-volatile
 - Larger capacity
 - Slower than primary memory

Tertiary and Archival Storage

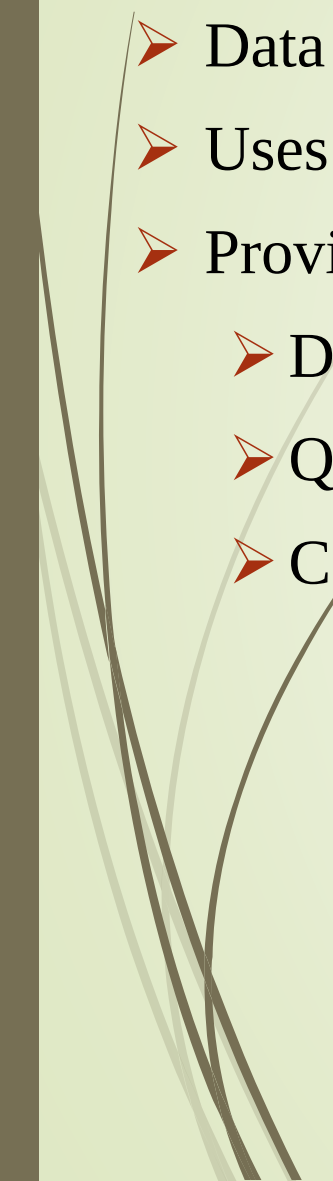
- Used for **backup and long-term archiving**
- Examples:
 - Magnetic tapes
 - Optical disks
- Low cost per unit storage
- Slow access time

Data Storage Architectures:-

File-Based Storage Systems

- Data stored as files in directories
- Managed by operating systems
- Simple and widely used
- **Limitations :-**
 - Poor scalability
 - Limited support for concurrent access
 - Data redundancy issues

Database Storage Systems



- Data stored in structured form using databases
- Uses schemas, tables, and records
- Provides:
 - Data integrity
 - Query support
 - Concurrency control

Distributed Storage Systems:-

- Data stored across multiple machines
- Provides:
 - Fault tolerance
 - Scalability
 - High availability
- **Examples :-**
 - Distributed file systems
 - Cluster-based storage

Data Storage Models:-

Centralized Storage

- Data stored at a single location
- Easy to manage
- Single point of failure

Distributed Storage

- Data distributed across multiple nodes
- Improves:
 - Reliability
 - Performance
- Requires coordination mechanisms
- **focus:**
Trade-off between **consistency, availability, and fault tolerance.**

The world is becoming increasingly data-driven. In fact, it's estimated that for every person on earth, 1.7 megabytes of data is created every second.



Data processing:-

- It's the conversion of data and raw information into a desirable, usable, and understandable format, usually over multiple steps.
- Data processing steps include collecting, recording, organizing, structuring, storing, retrieving, using, and disseminating.
- Processed data may take the form of an image, graph, table, chart, or any other format.

Why is data processing important?

- Helps information become more reliable and accurate.
- Helps store, distribute, and report on information.
- Enables better analysis and presentation of data.
- Helps businesses be productive, reduce costs, and increase profits.

What are the steps involved in data processing?

1. Data gathering
2. Data storage
3. Data preparation
4. Data processing
5. Data analysis
6. Data presentation

6 STAGES OF DATA PROCESSING

1

Data gathering

Collect trustworthy data from a variety of sources.

2

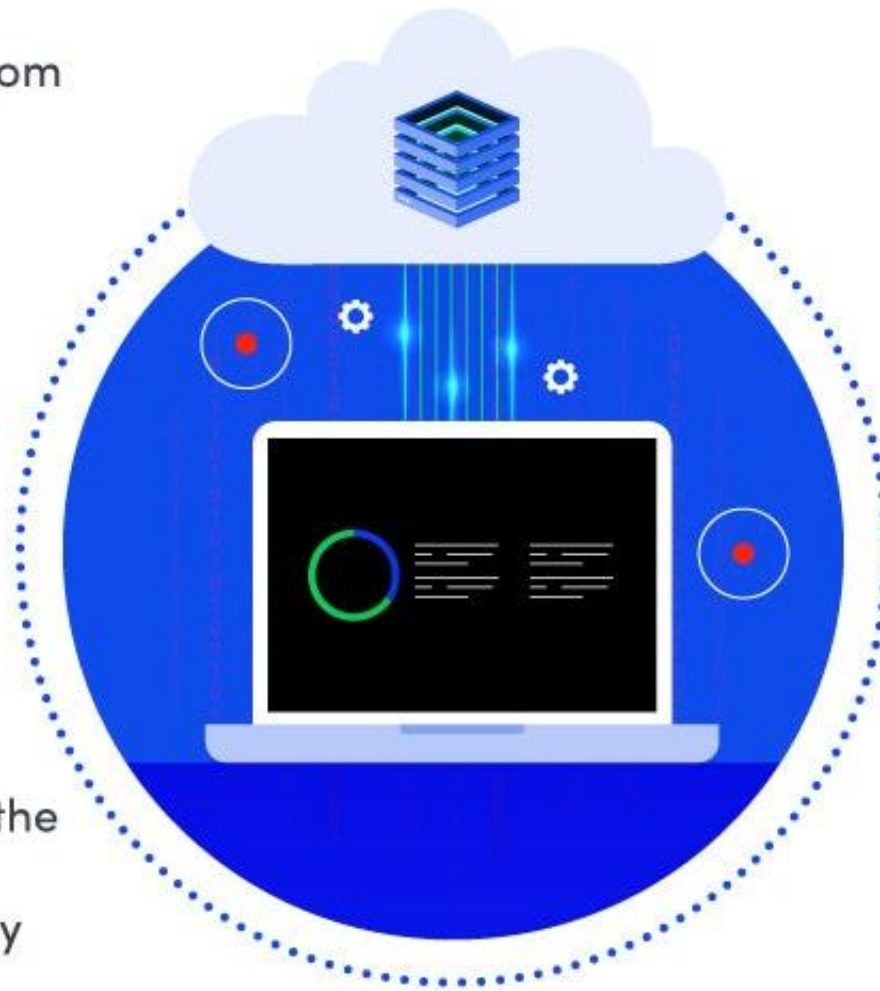
Data storage

Store the gathered data appropriately, in a data warehouse, lake or vault, for example.

3

Data preparation

Prepare the data for use, which involves evaluating the completeness, accuracy, timeliness and other quality metrics.



4

Data processing

Process the data, which may involve verification, transformation and organisation.

5

Data analysis

Analyse and interpret the data in the context of specific questions or concerns.

6

Data presentation

Present the data in a way that makes sense to the audience, including preparing tables, graphs or other visuals.